MULTI-ASSET QUANTITATIVE SWING TRADING SYSTEM

26-Feature Orthogonal ElasticNet ML + Markov-Switching GARCH Volatility Filter + 15D/20D Adaptive Runner & Dual-Bear Hedge

Universe: ~515 Liquid U.S. Equities (ex-UNH) + ETF Sleeve
Horizon: 7–15D Base / 20D Adaptive Runner (+1.5R / +3.0R / -1.0R)
Execution: Daily Post-Close (16:35 ET)

+192.0%	+270.1%	+110.9%	+80.9% / +56.2%	3.37 / 6.31	-23.6% / -31.6%
10-YR SWING SLEEVE CAGR	10-YR FULL CORE-SAT CAGR	40-YR GROSS CAGR	40-YR AFTER-TAX CAGR	10-YR SHARPE / SORTINO	MAX DRAWDOWN (10Y / 40Y)
Unlevered Equity + SGOV (2.79 SR)	With Dual-Bear Hedge (3.37 SR)	1985–2025 Continuous	TTS (28%) / NYC (51% Tax)	Equity-Only: 2.79 / 5.01	vs. QQQ -83.0% Max DD

1. EXECUTIVE THESIS & STRUCTURAL ALPHA EDGE

Why the Strategy Works Across Bull, Choppy & Crash Regimes

A. 26-Factor Orthogonal ML Selection

Every day after close, the engine scores **~515 liquid U.S. equities (ex-UNH)** across **26 orthogonalized predictors** (collinear duplicates pruned; scale-invariant `StdBeta = β·σ_X` attribution) to estimate the calibrated probability of hitting **+1.5R before -1.0R**.

B. 15D / 20D Adaptive Runner Payoff

Positions scale out 50% at **+1.5R** (locking stop to breakeven & +0.85R ratchet) and trail the remaining runner via a **2.5× ATR Chandelier Stop**. Confirmed mark-out leaders (`TrendQuality ≥ 0.65` & `RS_20 ≥ +5%`) extend from **Day 15 to Day 20** (+298.6 bps avg mark-out).

C. All-Weather Crisis Alpha Sleeve

Single-stock short signals suffer bull-market short squeezes, so individual equities are traded **strictly Long-Only**. In Bear Regimes (`QQQ < 50–lmMA` & `GLD` Bearish), idle cash rotates into index inverse ETFs (`75% FNGD + 25% SQQQ` or `SH / PSQ`) with a 15D time stop & -10% circuit into `SGOV`.

2. FIVE-STAGE QUANTITATIVE SIGNAL & EXECUTION ARCHITECTURE

Zero-Lookahead Purged Walk-Forward Pipeline

STAGE 1 • DATA
Universe Ingestion
Fetches adjusted OHLCV for ~515 liquid equities (ex-UNH) + `SPY`, `QQQ`, `GLD`, & `^VIX`. Enforces $5M min daily dollar volume & 7-day earnings blackout.

STAGE 2 • FEATURES
26 Orthogonal Factors
Computes `CMF_20D`, `RS_60`, `EfficiencyRatio_20D`, `Prox52WHigh`, `IdioAlpha_20D`, `RangeToCloseVol_20D`, & `LowerWickDemand_10D`.

STAGE 3 • REGIME
MSGARCH + Macro Gate
Classifies market into **Bull**, **Defensive/Choppy**, or **Dual-Bear** using QQQ/GLD 50-day `lmMA` slope, VIX, and 2-state MS-GARCH probability.

STAGE 4 • ML SCORING
ElasticNet + Platt
Purged 5-fold CV ElasticNet (`α=0.5` , 5-bar embargo) + Platt scaling predicts calibrated *P*(Up) and ranks via upgraded `Quality_Score`.

STAGE 5 • SIZING
Vince Optimal *f*
Sizes top-3 candidates (1.5x margin cap, 8% portfolio heat budget, max 2–3/sector) and deploys uninvested cash into `SGOV` or Dual-Bear Hedge.

Macro Regime Gate & Asymmetric Hybrid Probability Cutoffs

Market Regime	Trigger Condition	Min Win Prob	Sleeve Action
Bull Regime	`QQQ ≥ 50–lmMA` & Slope > 0, `VIX < 22`	**≥ 58.0%**	Up to 3 Long Swings + `QQQ` / `SGOV` Cash
Defensive (Leader)	**Defensive Gate & Stock RS_{20} ≥ +5.0%**	**≥ 60.0%**	**Max 2 Slots • Top-RS Leaders Only**
Defensive (Base)	`QQQ < 50–lmMA` or MS-GARCH High-Vol	**≥ 65.0%**	Max 2 Slots • High-Conviction + `GLD` / `SGOV`
Dual-Bear Hedge	Both `QQQ` & `GLD` < `50–lmMA` (Bear)	**65% / HEDGE**	Idle Cash → `75% FNGD + 25% SQQQ` (15D/-10% Stop)

Upgraded Cross-Sectional Ranker: Qualifying `BUY` candidates (with $RS_{20} \geq 0$ & $RS_{60} > -5\%$) are ranked by $Quality_Score = P(Up) \times (1 + RS_{20} + 0.20 \cdot RS_{60} + 0.5 \cdot SecSpread + AsymBoost + 0.25 \cdot (R^2 - 0.50) + 0.20 \cdot (ER_{20} - 0.30) + 0.08 \cdot IdioAlpha_{20D})$.

26-Feature Orthogonal Continuous Factor Taxonomy (7 Collinear/Step Flags Replaced)

- **Trend Linearity, Path Efficiency & Breakout Proximity (6 features):** `TrendQuality` (20d `lmMA` R^2), **`EfficiencyRatio_20D`** (Kaufman net-move / path-length ratio; IC = +0.0186), **`Prox52WHigh`** (252-day high proximity; 114.4 bps spread), `SlopeFast`, `SlopeSlow`, and `ADX_14`.
- **Regime Volatility & Contraction (6 features):** `GARCH_Vol`, `GARCH_VolPct`, `MSGARCH_ProbHighVol` (2-state Markov-Switching GARCH + logistic regime fallback), `AsymVolRatio`, `VolSqueeze`, and `SqueezeRelease`.
- **Microstructure, Chaikin Money Flow & Wick Geometry (7 features):** **`CMF_20D`** (#1 overall |β| = 0.0102; 20d volume-weighted CLV pullback washout, 132.1 bps spread), **`LowerWickDemand_10D`** (10d ATR-normalized lower-shadow buying), **`RangeToCloseVol_20D`** (Parkinson High-Low / Close-to-Close stop-hunt guard), `CLV_5D`, `IntradayVsGap_10D`, `VolumeRatio`, and `OBV_Slope`.
- **Multi-Horizon Relative Strength & Residual Alpha (7 features):** **`IdioAlpha_20D`** (20d beta-adjusted residual Information Ratio vs. SPY), **`RS_60`** (60d quarterly structural RS vs. SPY; +58.1 bps spread), `RS_20`, `SectorSpread` (13-sector ETF mapped), `DistPct`, `ZScore`, `RSI_14`, and `MACD_Hist`.

3. POSITION SIZING, PORTFOLIO HEAT CONTROLS & ADAPTIVE RUNNER LIFECYCLE

Strict Downside Truncation with 20-Day Right-Tail Convexity

Lifecycle Stage / Control	Quantitative Rule / Trigger	Execution Mechanics & Institutional Purpose
Entry & Position Sizing	**Limit Buy** • Vince Optimal *f* / Inverse-Vol	Allocates across up to 3 concurrent swing positions (2 in Defensive regime) using 1.5x margin buying power and **3.0x ATR overextension filter**.
Portfolio Heat & Circuit Breaker	**8.0% Max Open Risk • -12.0% Drawdown CB**	Total risk across active stops cannot exceed 8.0% of equity. If account drawdown reaches -12.0% from peak equity, position sizing automatically throttles by 50%.
Initial Downside Stop	**Hard Stop at -1.0R** ($Entry - 2.0 \cdot \sigma_{daily}$)	Dynamic volatility-scaled stop placed immediately upon entry (~-3.5% to -5.5% depending on GARCH conditional vol).
Tier 1 Scale-Out & Ratchet	**Sell 50% at +1.5R ($Entry + 3.0 \cdot \sigma_{daily}$)**	**Banks profit on half the position, moves stop-loss on the remaining 50% to Breakeven, and activates a +0.85R profit-lock floor once price reaches +2.2R.**
Adaptive Runner Horizon	**Day 15 Base → Day 20 Leader Extension**	**Trails the runner via $High_{max} - 2.5 \cdot ATR_{14}$. If `tier1_filled`, `TrendQuality ≥ 0.65`, $RS_{20} \geq +5\%$, and gain ≥ +2%, extends time barrier from Day 15 to Day 20.**
Cash & Dual-Bear Hedge Sleeve	**QQQ / GLD / SGOV + 75% FNGD / 25% SQQQ**	Uninvested cash rotates into `QQQ` (Bull), `GLD` (Defensive Gold Bull), or `75% FNGD + 25% SQQQ` (Dual-Bear, capped at 15 days or -10% max loss into `SGOV`).

Repository: `github.com/cmejo/R_SwingTrading` • Core Engine: `daily_signal.R` | `R/04_feature_pipeline.R` | `R/05_logistic_model.R`

HISTORICAL RETURN AUDIT (1985–2026) & TURNKEY DEPLOYMENT

10-Yr Upgrade Attribution (2016–2026), 40-Yr Continuous Tax Audit (1985–2025), 5 Non-Selective 10-Yr Epochs & Execution Guide

Friction: 10 bps Round-Trip Slippage Included
Tax Audit: Gross vs. Sec. 475 TTS (28.4%) vs. NYC (51.2%)
Benchmarks: SPY, QQQ, VTI & 70/30 Blend

4. TEN-YEAR OUT-OF-SAMPLE WALK-FORWARD & UPGRADE ATTRIBUTION (2016–2026 • 2,520 TRADING DAYS)

161 Liquid Stocks (UNH Removed)

Step-by-Step Model Evolution (2016–2026 Out-of-Sample)	Equity Swing CAGR (No Inverse ETFs)	Swing Max DD	Swing Sharpe / Sortino	Swing Calmar	Full Core-Sat CAGR (With Hedge)	Full Sharpe / Sortino	Full Calmar	Win Day %
1. Original 19-Feature Baseline (With UNH, 162 Stocks)	+102.91%	-31.05%	1.86 / 3.07	3.31	+167.54%	2.53 / 4.40	5.69	54.32%
2. Prior 26-Feature Model (With UNH, 162 Stocks)	+137.82%	-30.97%	2.23 / 3.79	4.45	+210.79%	2.87 / 5.11	7.16	55.46%
3. Prior 26-Feature Model (UNH Removed, 161 Stocks)	+137.62%	-31.62%	2.23 / 3.79	4.35	+210.53%	2.87 / 5.11	7.15	55.50%
4. + Stage-1 Upgrades (4 Orthogonal Feats + 20D Runner, 161 Stocks)	**+178.53%**	**-25.31%**	**2.67 / 4.74**	**7.05**	**+254.04%**	**3.26 / 6.02**	**8.15**	**56.36%**
5. LATEST Stage-2 System (+ CMF_{20}, RS_{60}, ER_{20} & Sector Fix)	**+192.02%**	**-23.61%**	**2.79 / 5.01**	**8.13**	**+270.12%**	**3.37 / 6.31**	**8.66**	**56.32%**

Year-by-Year Pure Equity Swing Sleeve Returns (Latest Upgraded vs. QQQ): **2017:** +37.7% (QQQ +32.7%) • **2018:** +14.4% (QQQ -0.1%) • **2019:** +125.9% (QQQ +39.0%) • **2020:** +229.3% (QQQ +48.4%) • **2021:** +177.3% (QQQ +27.4%) • **2022 Bear:** +72.6% Swing / +165.7% Full (QQQ -32.6%) • **2023:** +185.6% (QQQ +54.9%) • **2024:** +443.8% (QQQ +25.6%) • **2025:** +406.7% (QQQ +20.8%) • **2026 YTD:** +455.0% (QQQ +16.8%).

5. FORTY-YEAR CONTINUOUS BACKTEST & AFTER-TAX REGIMES (1985–2025)

Annual Tax Drag Paid Out of Equity Every Dec 31

Portfolio / Tax Jurisdiction (1985–2025)	CAGR	Ann. Vol	Sharpe	Sortino	Max DD
Strategy — Gross (IRA / Roth / PR Act 60)	**+110.91%**	**28.53%**	**2.76**	**4.59**	**-31.62%**
Strategy — Trader Tax Status (28.4% Tax)	**+80.86%**	**28.23%**	**2.01**	**2.98**	**-37.29%**
Strategy — NYC Resident (51.2% Top Tax)	**+56.23%**	28.94%	1.36	1.68	-50.45%
Nasdaq-100 (`QQQ` Gross / NYC Net)	+14.33% / +12.91%	26.27%	0.65	0.89	-82.96%
Total US Market (`VTI` Gross / NYC Net)	+9.01% / +7.40%	17.56%	0.56	0.74	-56.65%
70/30 `VTI/QQQ` Blend (Gross / NYC Net)	+10.83% / +9.26%	19.37%	0.62	0.83	-61.35%

Calendar Decade Breakdown	Strategy CAGR	Strat Max DD	Strat Sharpe	QQQ CAGR (Max DD)
1985–1994 (Pre-Tech Dawn)	**+94.64%**	-23.69%	2.67	+14.59% (-39.9%)
1995–2004 (Dot-Com Bubble/Bust)	**+114.49%**	-24.66%	2.74	+14.61% (-83.0%)
2005–2014 (GFC & Recovery)	**+77.97%**	-31.62%	2.19	+9.98% (-53.6%)
2015–2024 (Tech & COVID Era)	**+152.05%**	-21.27%	3.33	+17.38% (-35.6%)
2025–2026 (Recent Walk-Forward)	**+177.37%**	**-18.81%**	**3.23**	**+19.47% (-22.9%)**

Capacity & Liquidity Scaling ($10k vs. $10M+): At account sizes of **$10k–$2M**, the full system compounds at **100%+ Gross CAGR**. Above **$5M–$15M+**, market-impact caps and unhedged large-cap constraints compress returns toward **25%–36% CAGR**. Cap single execution sleeves at $1M–$2.5M.

6. FIVE NON-SELECTIVE 10-YEAR HISTORICAL EPOCHS ($10,000 FRESH START PER EPOCH)

Stress-Tested Across Dot-Com Bust, Lost Decade & 2008 GFC

10-Year Historical Epoch ($10,000 Initial Capital)	Gross CAGR	Gross Final Wealth	TTS Net CAGR (28.4% Tax)	NYC Net CAGR (51.2% Tax)	Strategy Max DD	Strategy Sharpe	QQQ Gross CAGR	QQQ Max DD
Epoch 1: Dot-Com Boom & Crash (1995–2004)	**+114.49%**	**$16,913,925**	+83.01% ($3.46M)	+57.39% ($766k)	-24.66%	2.74	+14.61% ($32.2k)	-82.96%
Epoch 2: The Equity "Lost Decade" (2000–2009)	**+90.77%**	**$5,193,144**	**+66.21% ($1.31M)**	**+45.97% ($359k)**	**-25.13%**	**2.45**	**-6.70% ($4.1k)**	**-82.96%**
Epoch 3: Global Financial Crisis (2007–2016)	**+69.29%**	**$1,580,450**	+50.51% ($490k)	+35.04% ($166k)	-31.62%	2.01	+10.63% ($22.6k)	-53.55%
Epoch 4: QE Bull to COVID Shock (2012–2021)	**+114.82%**	**$17,074,231**	**+83.26% ($3.49M)**	**+57.55% ($772k)**	**-24.74%**	**2.86**	**+21.73% ($58.6k)**	**-28.56%**
Epoch 5: Pre-Tech Modernization (1988–1997)	**+97.50%**	**$7,610,162**	+70.43% ($1.73M)	+48.48% ($435k)	-24.66%	2.56	+19.97% ($51.2k)	-32.93%

7. TURNKEY OPERATOR GUIDE: HOW OTHERS CAN DEPLOY & RUN THE STRATEGY

10-Minute Setup on macOS / Linux

Step 1: Environment & Initialization

- **Requirements:** R (4.2+) with `quantmod`, `TTR`, `glmnet`, `MSGARCH`, `data.table`, `isonlite`; Python (3.10+) with `schwab-py` / `ib_insync`, `yfinance`.
- **Clone Repository:**
  `git clone https://github.com/cmejo/R_SwingTrading.git`
- **Set Starting Account Equity:**
  `python3 reset_evaluation.py --cash 25000`
  Initializes `portfolio.json` and sets a clean benchmark baseline against `SPY`.

Step 2: Daily Execution (16:35 ET)

- **Mode A — Semi-Automated Ticket Workflow:**
  1. Run `Rscript daily_signal.R` at 16:35 ET.
  2. Open `LATEST_TICKET.txt` for exact share counts, limit prices, 3-tier stop ratchets, and `SGOV` / `FNGD` / `SQQQ` hedge adjustments.
  3. Log fills via `python3 update_portfolio.py`.
- **Mode B — Automated Broker API:**
  Authenticate Schwab (`execute_broker.py --broker=schwab --auth`) or IBKR and schedule `run_daily.sh --broker=schwab --live` via cron at `35 16 * * 1-5`.

Step 3: Monitoring & Governance

- **Live HTML Dashboard:** Run `python3 track_performance.py` to generate `performance_dashboard.html` with real-time quotes, open positions, hedge sleeve weights, and cumulative alpha vs. SPY.
- **Statistical Review Cadence:**
  • *Daily:* Verify cron & stop-loss integrity.
  • *30 Closed Trades (~3–4 Mos):* Check win-rate (55%–62%) & payoff ratio (≥1.4x).
  • *6–12 Months:* Evaluate Sharpe (≥1.5) & Max DD (≤28%).

Important Risk Disclosure: Historical backtest and out-of-sample walk-forward results include 10 bps round-trip slippage and realistic borrow/ETF expenses, but do not guarantee future performance. Leveraged inverse ETFs/ETNs (`FNGD`, `SQQQ`) experience path-dependent volatility decay if held without time/loss stops; the strategy enforces a strict **15-day time stop and -10.0% max-loss circuit into SGOV**.